TEXORA · LECTURE SHEET

Project Plan for Entire Weaving Shed

Sheets 1–5 calculated one process at a time. A project plan chains them: from one order — a fabric, a quantity, a date — it derives yarn, beams, size, machines, space and people, each figure feeding the next. This sheet works one complete case from order to shed layout.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 6 · SPECIALIZATION — FABRIC MANUFACTURING

1. The order and the cloth

ITEM	VALUE
Order	100,000 m of grey plain-weave cotton fabric, delivery in 20 working days
Grey width	58 in (1.473 m); reed width 60 in
Construction	EPI 88, PPI 64; warp and weft Ne 24 (24.6 tex)
Total ends	88 × 60 = **5,280 ends**
Warp crimp + contraction allowance	9% (warp length per metre of fabric = 1.09 m)

2. Step 1 — Yarn requirement

WORKED CALCULATION — WARP AND WEFT TOTALS

Warp per metre of fabric = 5,280 ends × 1.09 m × 24.6 tex ÷ 1,000 = **141.6 g**.

Total warp = 141.6 g × 100,000 m = **14,160 kg**.

Picks per metre = 64 × 39.37 = 2,520; pick length 1.55 m (reed width + waste).

Weft per metre = 2,520 × 1.55 × 24.6 ÷ 1,000 = **96.1 g**; total weft = **9,609 kg**.

Total yarn = 14,160 + 9,609 = **23,769 kg**. (Check by area: 237.7 g/m ÷ 1.473 m width = GSM 161 g/m² — the cloth's own specification, closing the loop with Sheet 5's method.)

3. Step 2 — Looms

WORKED CALCULATION — LOOM FLEET

Loom at 200 ppm, efficiency 88%, 64 PPI: production = 200 × 60 × 0.88 ÷ 64 ÷ 39.37 = **4.19 m/h**.
Working day = 22 h (three shifts less changeovers): 4.19 × 22 = **92.2 m per loom per day**.
Looms = 100,000 ÷ (92.2 × 20) = 54.2 → **55 looms**.

4. Step 3 — Preparation: beams, size, winding

- **Warping:** creel of 528 cones (Sheet 2) divides the warp exactly: 5,280 ÷ 528 = **10 section beams per set**, no short beam.
- **Weaver's beams:** beam length 2,500 m of fabric (warp length 2,725 m): 100,000 ÷ 2,500 = **40 beams** over the order — 2 beams per day to size and deliver. Each set-beam of 528 ends × 2,725 m × 24.6 tex weighs 35.4 kg (Sheet 2, Practice 2), so a full set of ten weighs 354 kg at the sizing creel.
- **Sizing:** add-on 10% on warp (Sheet 3): dry size = 14,160 × 0.10 = **1,416 kg of solids** for the order; at the Sheet 3 recipe cost of Tk 159/kg that is ≈ Tk 225,100 of size material.
- **Winding:** all 23,769 kg passes the winders. At 1,100 m/min, 80% efficiency, 22 h/day: per spindle per day = 1,100 × 60 × 22 × 0.80 × 24.6 ÷ 10^6 = **28.58 kg**. Spindle-days needed = 23,769 ÷ 28.58 = 832; over 20 days → 41.6 → **42 spindles** — install one 48-spindle frame (Sheet 1's rounding rule).

5. Step 4 — Space and layout

Floor allowance per installed loom, including its share of aisles, is 7.5 m²: the loom hall needs 55 × 7.5 ≈ **413 m²**. Preparation (winding, warping, sizing) is laid out in the yarn-flow order so warp moves forward, never back; the cloth room sits at the loom hall's exit. The layout below is schematic — its job is to prove the flow and the areas, not to fix a building.

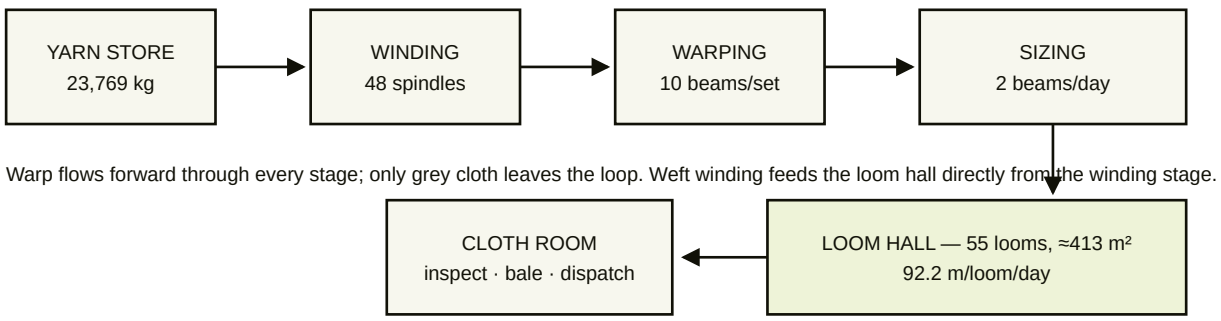


Figure 1 — Shed layout as a flow: store → winding → warping → sizing → looms → cloth room.

6. Step 5 — People and the plan table

STAGE	CAPACITY FIGURE	INSTALLED	CHECK
Winding	28.58 kg/spindle/day	48 spindles	48 × 28.58 × 20 d = 27,437 kg > 23,769 kg ✓
Warping	10 section beams/set	1 sectional warper	2 sets/day needed; a set winds in well under a shift ✓
Sizing	2 beams/day	1 sizing machine	2,725 m beam ≈ 75 min at 45 m/min, 80% — light duty ✓
Weaving	92.2 m/loom/day	55 looms	55 × 92.2 × 20 = 101,420 m > 100,000 m ✓

Manpower follows the machine tending norms of the mill (weavers per loom set, winders per frame, one sizing crew per shift); the plan fixes the *machine* numbers first because people are hired to machines, not to metres. The plan's last discipline is the margin column: every stage must show capacity above the order's demand, because the stage with no margin becomes the shed's true speed.

Glossary

Project plan — the chained calculation from order to machines, materials, space and people. **Set** — section beams combining into one weaver's beam. **Beam length** — fabric metres woven from one weaver's beam. **Capacity check** — verifying each stage's installed capacity exceeds the plan's demand on it.

Key points

• Chain of the plan: order → yarn (23,769 kg) → looms (55) → preparation (40 beams, 1,416 kg size, 42→48 spindles) → space (413 m² loom hall) → people.
• Choose creel capacity to divide the total ends exactly (528 into 5,280 = 10 full beams).
• Every stage carries a written margin check; the tightest margin is the shed's real capacity.
• Sizing material for the order: 1,416 kg solids ≈ Tk 225,100 at the Sheet 3 recipe.

Practice

1. The order doubles to 200,000 m in the same 20 days. Which plan figures double exactly, and which do not?
2. Recompute the loom fleet if efficiency falls to 82%.
3. Why was a 48-spindle winder installed when the calculation said 42?
4. The sizing machine can size 6 beams/day. How many days of sizing does the order need, and what does that imply for when sizing must start relative to weaving?

Answer key

1. Yarn, beams, size solids and spindle-days double exactly (they are proportional to metres). Looms double too at fixed days (110). What does *not* double: the number of machines with spare capacity already in hand (the warper and sizer still serve alone), the cloth room's peak, and the margins — each must be rechecked, not scaled by habit.
2. Production = 200 × 60 × 0.82 ÷ 64 ÷ 39.37 = 3.91 m/h → 85.9 m/day; looms = 100,000 ÷ (85.9 × 20) = 58.2 → **59 looms**. Six points of efficiency cost four looms.
3. Winders are bought as complete frames; 42 spindles is not a purchasable unit, and the 6 spare spindles absorb breakdowns and count changes (Sheet 1's rounding rule).
4. 40 beams ÷ 6 per day = **6.7 days** of sizing inside a 20-day order: sizing runs well ahead of the looms' consumption (2 beams/day), so it need not start before weaving — but the first beams must exist before the first looms are gaited, so sizing starts first by the gaiting lead time, then paces at 2 beams/day.

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